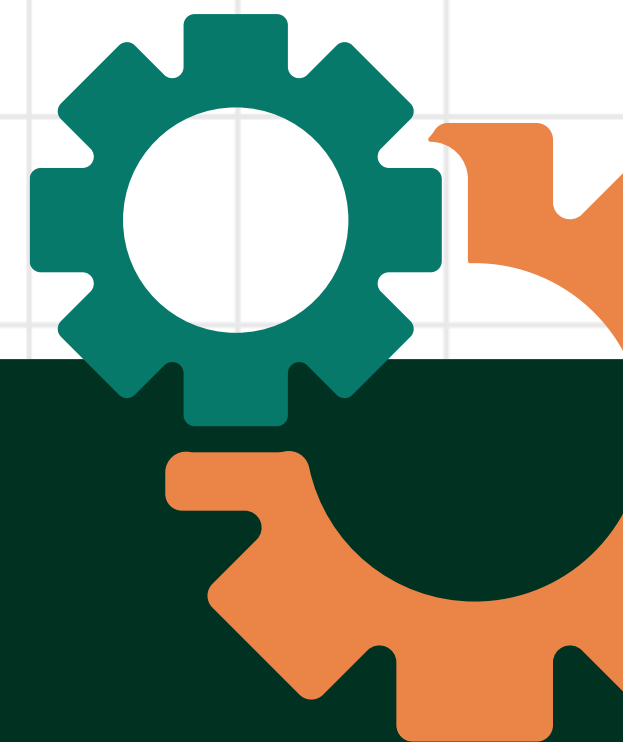


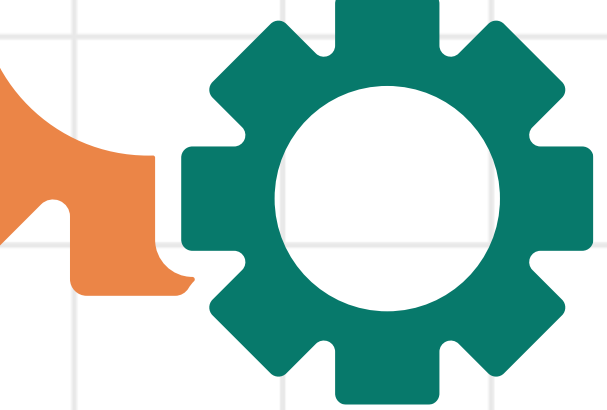


GENERAL SCIENCE
WEEK 3

SIMPLE AND COMPOUND MACHINES

LUIS ANTHONY N. GRADO

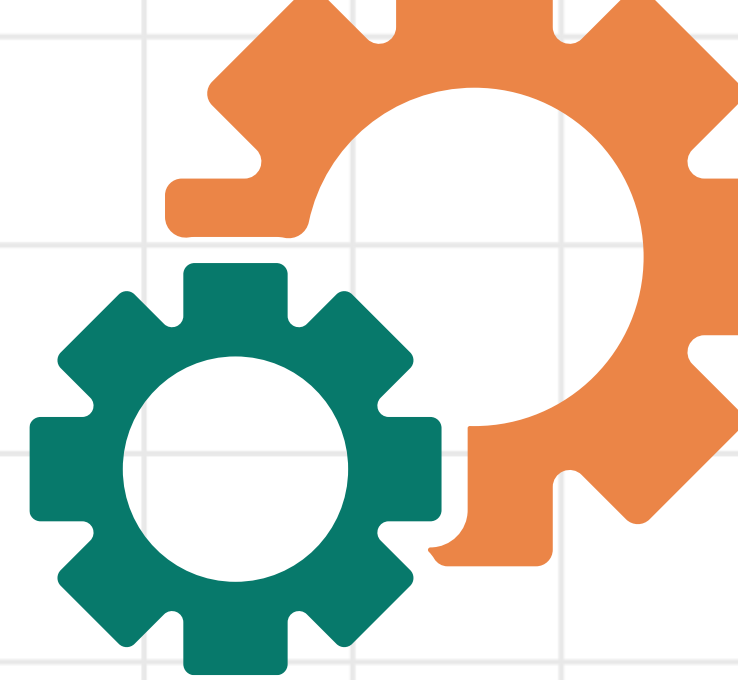




DAY 1: LEARNING OBJECTIVES

- Define simple and compound machines and explain how they differ in structure and function.
- Explain how machines convert straight-line (translational) movement into spinning (rotational) movement, or vice versa.
- Identify the specific forces that act on a machine during operation.

RECAP:



Machine Example

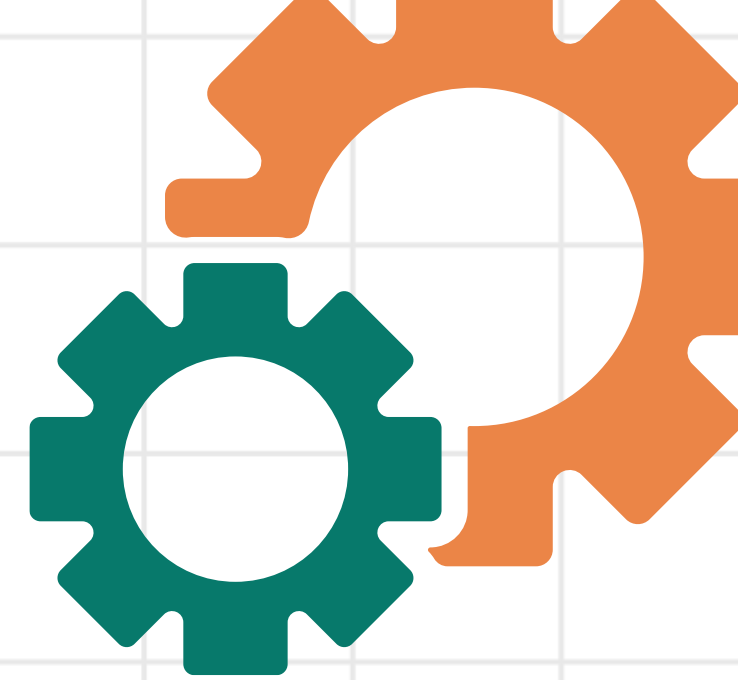
Example: Bicycle

Input Motion (User)

Rotational (Pedaling)

Output Motion (Machine)

Translational (Moving Forward)



RECAP:

- What are examples of machines you use every day?
- How do they function in terms of motion?



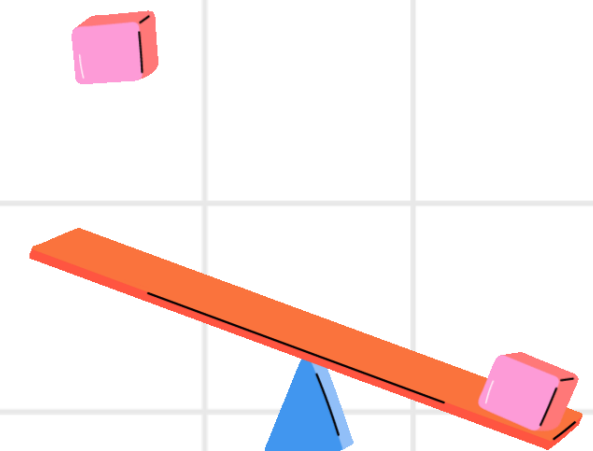
Do you notice how the machine often changes the type of motion?
A bicycle takes your legs going in circles and turns it into a straight line.
That change is exactly what simple and compound machines are designed to do.

GROUPWORK

INSPIRE INNOVATION: Comparison from Old to Present Designs
In your assigned groups, you will become "Mechanical Historians." You are going to investigate how machine designs have changed over time and identify the physics that stayed the same.

1. Join Your Group: Move to your designated station based on the machine type your group was assigned:

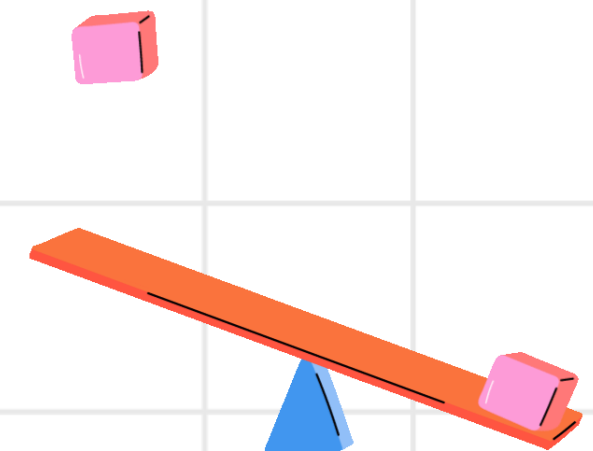
- Group 1: Bicycle Design
- Group 2: Wind Turbines
- Group 3: Cranes and Lifting Machines
- Group 4: Cars and Engines



GROUPWORK

INSPIRE INNOVATION: Comparison from Old to Present Designs

Review the Case Study: Read through the "Old vs. Present" design handouts provided. Look closely at the diagrams of the vintage models compared to the modern versions.

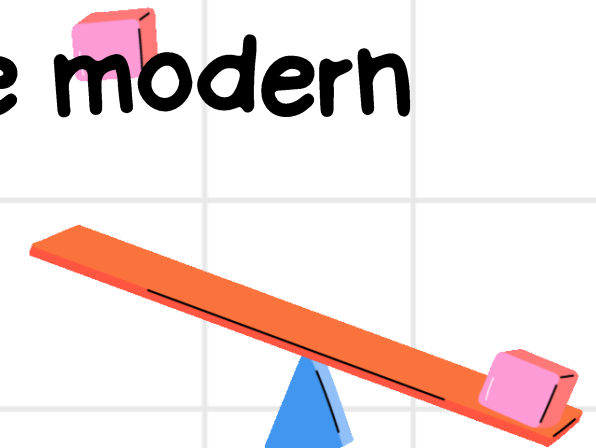


GROUPWORK

INSPIRE INNOVATION: Comparison from Old to Present Designs

Guide Questions:

1. How do old and new machine designs use rotational and translational motion differently?
2. How do modern machines convert rotational motion into translational motion more efficiently?
3. What improvements in motion mechanics make modern machines better than older ones?

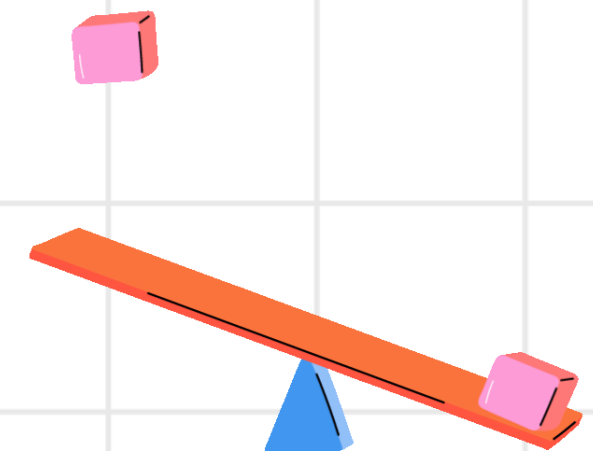


GROUPWORK

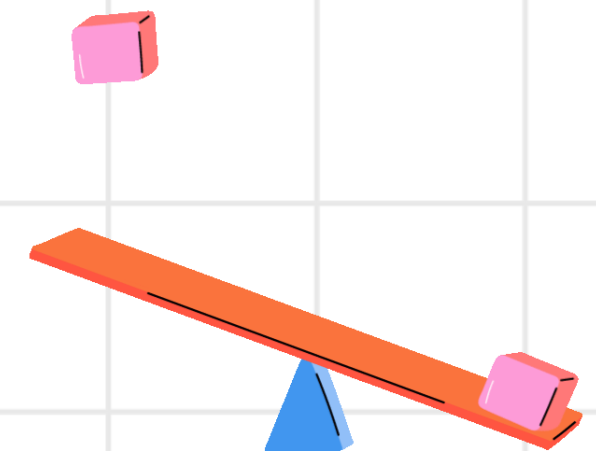
INSPIRE INNOVATION: Comparison from Old to Present Designs

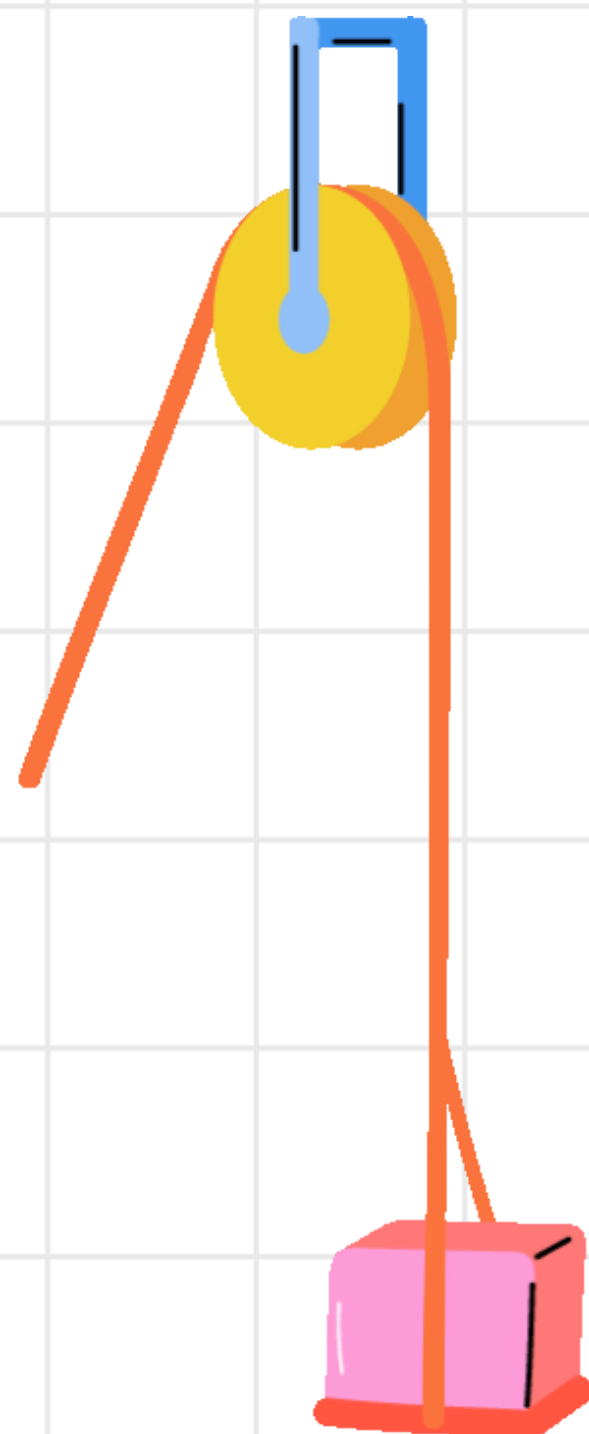
PRESENTATION

4-5 MINS/GROUP



“Every great invention started as a simple push or pull; understanding these motion principles allows you to look at the machines of today and creatively rethink how they might solve the challenges of tomorrow.”





**WHAT IS A
MACHINE?**

NETFLIX

MAC



QUESTIONS:

1. According to Rancho, what is the simplest definition of a machine?
2. What examples did Rancho use to prove his point during the scene?

MACHINE

**"Anything that reduces human effort or makes
work easier/faster."**

-RANCHO (3 Idiots)

2 CATEGORIES OF MACHINE

SIMPLE

COMPOUND

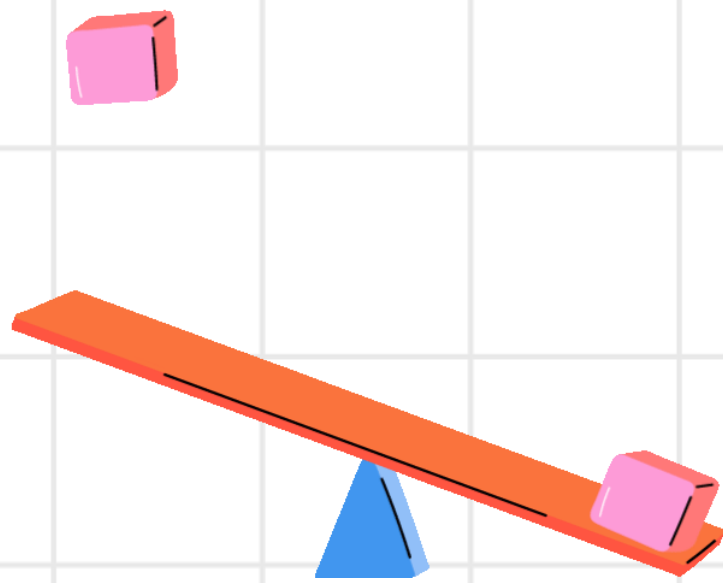
SIMPLE MACHINE

A simple machine is a mechanical device that changes the direction or magnitude of a force using a single, basic mechanism. They are the fundamental "building blocks" of all other machines.

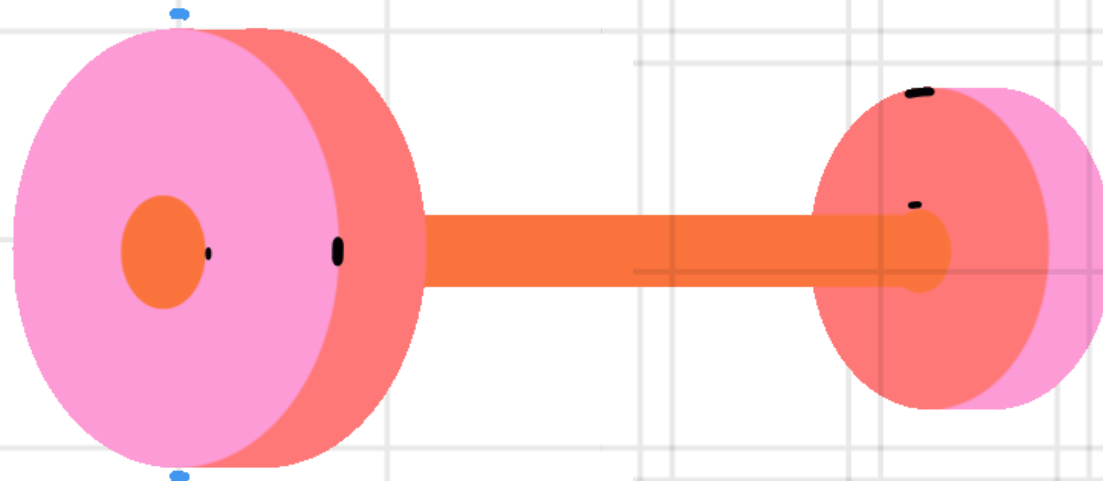
- How they work: They typically use a single applied force (Effort) to overcome a single resistance (Load).
- The Trade-off: Most simple machines provide Mechanical Advantage, meaning they allow you to use less force over a longer distance to move a heavy object.

SIMPLE MACHINES

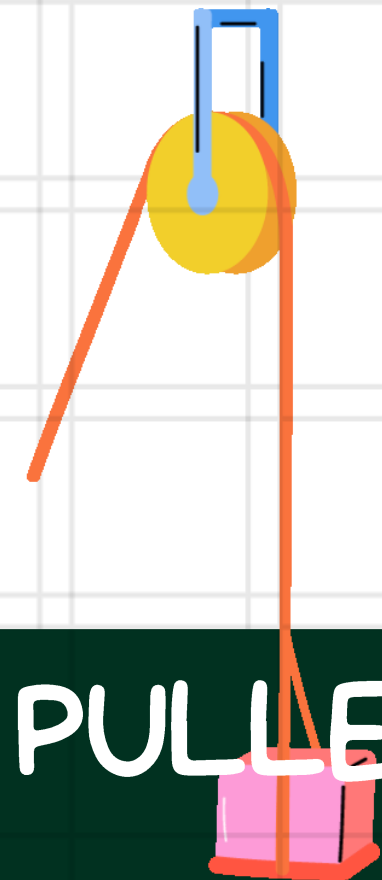
- The Six Classic Types:
 - Lever: A rigid bar pivoting on a fulcrum (e.g., a crowbar).
 - Wheel and Axle: A circular frame turning on a central shaft (e.g., a doorknob).
 - Pulley: A wheel with a grooved rim for a rope or chain (e.g., a flagpole hoist).



LEVER



WHEEL AND AXLE



PULLEY

SIMPLE MACHINES

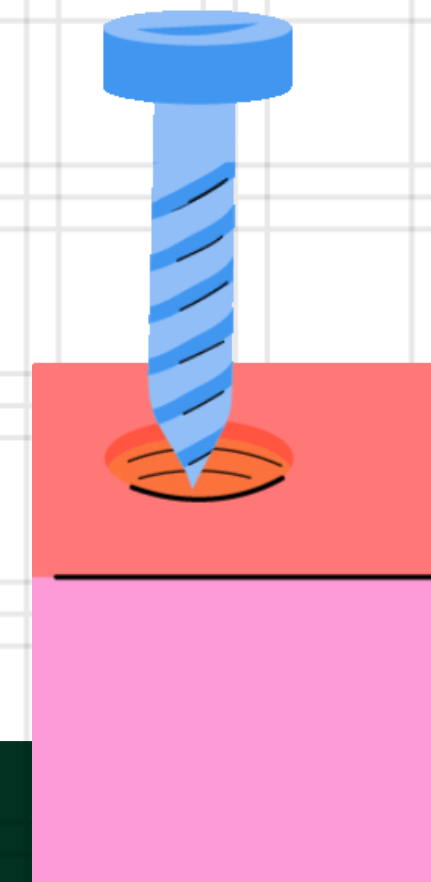
- The Six Classic Types:
 - Inclined Plane: A flat, sloped surface (e.g., a loading ramp).
 - Wedge: Two inclined planes joined back-to-back, used to split or lift (e.g., an axe).
 - Screw: An inclined plane wrapped around a cylinder (e.g., a jar lid).



INCLINED PLANE



WEDGE



SCREW

COMPOUND MACHINES

A compound machine (also known as a complex machine) is a device made up of two or more simple machines working together.

- **How they work:** In a compound machine, the output force of one simple machine often becomes the input force for the next. This creates a chain reaction of motion.
- **Motion Conversion:** Compound machines are excellent at converting motion, for example, converting the rotational motion of a gear into the translational (straight-line) motion of a cutting blade.

COMPOUND MACHINES

- Examples:
 - Bicycle: Combines wheels and axles, pulleys (the chain and sprockets), and levers (the handlebars and brakes).
 - Scissors: Combines two levers (the handles) and two wedges (the blades).
 - Can Opener: Uses a lever (handle), a wheel and axle (turning knob), and a wedge (the cutting wheel).



SIMPLE MACHINES: MOTION

MACHINE	PRIMARY MOTION	HOW IT WORKS
LEVER	ROTATIONAL	Pivots around a fixed point (fulcrum).
WHEEL AND AXLE	ROTATIONAL	A large wheel turns a smaller axle (or vice versa).
PULLEY	BOTH	Converts a linear pull (Translational) into a wheel spin (Rotational).
INCLINED PLANE	TRANSLATIONAL	Moving an object along a slope in a straight line.
SCREW	CONVERSION	Rotational (turning) is converted into Translational (moving in/out).
WEDGE	TRANSLATIONAL	A moving inclined plane that redirects a forward force into a sideways force

COMPOUND MACHINES: MOTION

A compound machine is just two or more simple machines working in a chain. They often act as a "motion relay race."

Example: A Bicycle

1. Pedals (Wheel & Axle): Your legs move up and down (Translational), pushing the pedals in a circle (Rotational).
2. Gears & Chain (Pulley/Sprocket system): The rotation of the front gear is transferred via the chain to the back gear.
3. Rear Wheel (Wheel & Axle): The rotation of the gear spins the wheel.
4. The Result: The spinning wheel pushes against the ground, moving the entire bike forward in a straight line (Translational).

FORCES ACTING ON A MACHINE

WEIGHT

TENSION

TORQUE

FRICTION

FORCES ACTING ON A MACHINE

WEIGHT

Weight is the gravitational pull on an object. In the world of machines, weight is almost always the Load; the thing you are trying to lift, move, or hold.

- How it acts: It always pulls straight down toward the center of the Earth.
- Machine interaction: On an inclined plane or wedge, weight is what creates the "push-back" you feel. On a lever, the weight of the object on one end is what determines how much force you need to apply to the other end.

FORCES ACTING ON A MACHINE

TENSION

Tension is a pulling force that travels through strings, ropes, cables, or chains. It is the "messenger" that carries force from one part of a machine to another.

- How it acts: It pulls equally in both directions along the length of the rope or cable.
- Machine interaction: This is the primary force in pulleys. When you pull a rope, tension is what actually lifts the weight. In compound machines like a bicycle, tension in the metal chain transfers the power from your pedals to the back wheel.

FORCES ACTING ON A MACHINE

TORQUE

Torque is all about rotation. It is "force with a handle."

- How it acts: It is a twisting force. The amount of torque depends on how hard you push AND how far you are from the pivot point (the fulcrum).
- Machine interaction:
 - * In Levers: When you push down on a crowbar, you are creating torque to pry something open

FORCES ACTING ON A MACHINE

FRICTION

Friction is the force that resists motion when two surfaces touch. It is the "tax" you pay on every movement.

- How it acts: It always acts in the opposite direction of the way the machine is trying to move.
- Machine interaction:
 - * In Screws: Friction is actually helpful; it's what keeps a screw from spinning back out of the wood.

QUIZ!

1. Which machine is used to split things apart (like an axe)?

A) Lever

B) Wedge

C) Pulley

D) Wheel

2. A spinning wheel moves a car forward in a straight line. This is turning:

A) Spinning motion into Straight motion

B) Straight motion into Spinning motion

C) Heavy motion into Light motion

D) Friction into Weight

QUIZ!

3. A screw is a machine that turns:

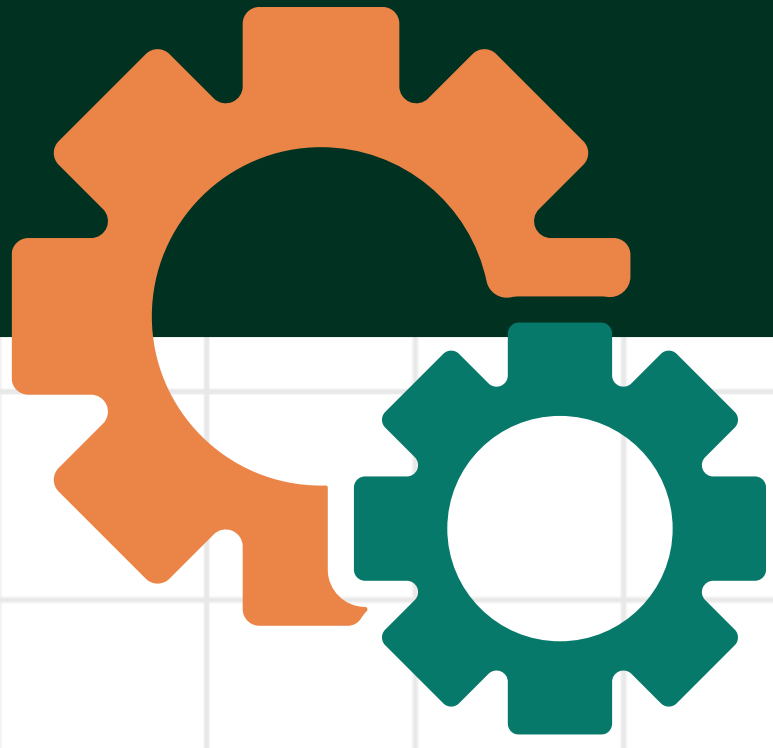
- A) Heat into motion
- B) Weight into friction
- C) A turning motion into a forward motion
- D) A pulling motion into a spinning motion

4. What do you call a machine that is made of two or more simple machines (like scissors)?

- | | |
|---------------------|---------------------|
| A) Simple machine | B) Compound machine |
| C) Electric machine | D) Fast machine |

QUIZ!

5. What is a machine? Define and give an example.



THANK YOU!

I hope you learn something new today.

